Lilyan Leblanc

+33 6 10 75 34 58 | contact@lilyan-leblanc.com | Full, clean driving license

Medical Robotics R&D Engineer | Medical Devices & Surgical Applications

Engineer with a PhD in robotics and 7+ years of experience across industry and academia, specializing in surgical robotics and medical devices. Experienced in C++/ROS 2, robotic prototyping, non-rigid 3D registration, calibration, and validation of complex systems in regulated R&D environments.

Technical Skills

- **Medical robotics software: C++, Git, MATLAB, ROS 2, Python**
- **Robotics: kinematics/dynamics modeling and identification, calibration, control**
- **3D medical imaging: non-rigid registration, surgical trajectory transfer**
- **Project & quality: Agile development, ISO 13485, QMS, CAPA, CSV, system validation**

Professional Skills

- **Critical thinking and problem-solving in complex technical environments**
- **Collaboration and teamwork in multidisciplinary settings**
- **Clear interpersonal communication across technical and non-technical stakeholders**
- **Cross-functional communication with engineering, clinical, quality and regulatory teams**

Professional Experience

2021-2025 | PHD RESEARCHER IN SURGICAL ROBOTICS | INSTITUT DES SYSTEMES INTELLIGENTS ET DE ROBOTIQUE - SORBONNE UNIVERSITE - FRANCE

- Developed robotic solutions for pedicle screw insertion (breach detection with patent filed, non-rigid registration) and contributed to surgical task execution software (C++, ROS 2, MATLAB)
- Designed non-rigid 3D medical image registration tools for surgical trajectory transfer
- Conducted experimental validations on ex vivo, cadaveric, in vivo, and animal models
- Participated in the FAROS (H2020) European research project in collaboration with multiple academic and industrial partners
- Taught courses in control theory (state-space, feedback control) and robotics (FK/IK, Jacobians, trajectory generation, impedance control)

2018-2021 | MEDICAL ROBOTICS ENGINEER | ZIMMER BIOMET ROBOTICS - FRANCE

- Designed, validated, and deployed automated calibration solutions for a 6-DOF Stäubli manipulator in the ROSA surgical robotic system (C++, VAL3), including kinematic parameters identification and surgical tools geometric calibration
- Automated calibration processes within an Agile team, contributing to a 15% reduction in manufacturing time
- Analyzed the error chain of the robotic system and conducted a state-of-the-art review of serial robot calibration methods
- Contributed to Computer System Validation, equipment and procedures validation, risk analysis, NCRs, and CAPAs (ISO 13485, 9001)

Education & Certifications

2021-2025 | PHD IN SURGICAL ROBOTICS | INSTITUT DES SYSTEMES INTELLIGENTS ET DE ROBOTIQUE - SORBONNE UNIVERSITE – FRANCE

2023 | UNIVERSITY DIPLOMA IN BIOLOGICAL AND MEDICAL ENGINEERING | SORBONNE UNIVERSITE - FRANCE

- Funding, protection, and translation of biomedical research into practical applications
- Entrepreneurship, funding and innovation strategy in a regulated biomedical environment

2016-2018 | MASTER OF SCIENCE IN IMAGING, ROBOTICS AND BIOMEDICAL ENGINEERING | TÉLÉCOM PHYSIQUE STRASBOURG - STRASBOURG UNIVERSITY - FRANCE

- Valedictorian. Coursework included Computer Science, Robotics, Control Systems, Computer Vision, Image Processing, and Signal Processing

2015-2018 | ENGINEERING DEGREE - INFORMATION TECHNOLOGY FOR HEALTHCARE | TÉLÉCOM PHYSIQUE STRASBOURG – STRASBOURG UNIVERSITY - FRANCE

- Valedictorian. Coursework included Computer Vision, Control Systems, Robotics, Computer Science, Signal Processing, Biomechanics, Electronics, Biology

Language Skills

- **French**: Native
- **English**: C1 – TOEIC 970/995
- **German** B1
- **Spanish**: A2

Patents and Publications

- 1 patent filed; 3 IEEE publications in medical robotics, 3D registration and real-time breach detection
- QR code to full list

Activities and Interests

- Advanced **scuba diver**, master **freediver**, **triathlete**
- **Cinema** and **theater**